

Internet Appendix for “The Intramonth Momentum Cycle”

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This Internet Appendix contains supplementary material for “The Intramonth Momentum Cycle.” It includes: (IA.1) data construction and sample formation, (IA.2) holding period analysis, (IA.3) transaction cost decomposition, (IA.4) T+1 settlement falsification tests, (IA.5) sliding window analysis, (IA.6) tail trimming analysis, (IA.7) crash-day concentration methodology, (IA.8) international evidence methodology and full country-level results, (IA.9) subperiod PreTOM concentration analysis, (IA.10) selling recency (fresh vs. stale losers), (IA.11) window dressing falsification, (IA.12) dividend cycle within-quarter detail, (IA.13) 52-week-high sort additional evidence, (IA.14) liquidity shock panel regression full specification, (IA.15) beta controls, (IA.16) T+3 to T+2 settlement transition, (IA.17) winner window effect, (IA.18) S&P 500 membership robustness, (IA.19) long-horizon reversion identifying the PreTOM-specific component, and (IA.20) additional figures.

IA.1 Data Construction and Sample Formation

The daily stock-level panel used in the main analysis is constructed as follows.

1. **CRSP input data.** We obtain the CRSP daily stock file from WRDS in Flat File Format 2.0 (CIZ). The file spans December 29, 1978 through December 31, 2025.¹

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¹Observations from 1978–1979 are retained in order to construct momentum signals for the first valid holding months in 1980.

2. **CRSP sample restrictions.** We restrict the CRSP sample to U.S. common operating-company equities listed on the NYSE, AMEX, or Nasdaq, and retain only regular-way, active observations.²
3. **Trading-day panel construction.** Using the CRSP trading calendar, we expand each security’s panel so that all trading days between its first and last observed trading day are included. Trading days without an observed CRSP record remain in the panel as missing rows. After applying the common-stock and exchange/trading-status filters, the sample contains 59,917,630 observations on 23,827 securities. Trading-day expansion increases this to 60,910,165 observations. Table IA.1 reports the corresponding attrition.
4. **Month-relative trading-day index.** Within each month, trading days are indexed so that the last trading day is labeled $T = 0$, the second-to-last is $T = -1$, and so on through $T = -9$ for the tenth-to-last trading day. Earlier trading days in the same month are indexed $T = 1, 2, \dots$
5. **Momentum construction.** For each stock-month, we compute 12-2 momentum from monthly returns aggregated from daily CRSP observations:

$$M_{i,m} = \prod_{s=m-12}^{m-2} (1 + r_{i,s}) - 1.$$

The resulting stock-month signal is assigned to all daily observations in holding month m .

6. **Momentum validity.** A momentum window is treated as valid only if it contains no trading-calendar-added empty rows and no missing-return indications according to the CRSP variable `DlyRetMissFlg`.³ We then retain only observations with valid

²In the CIZ format, these restrictions are implemented sequentially as follows: `ShareType='NS'`, `SecurityType='EQTY'`, `SecuritySubType='COM'`, `USIncFlg='Y'`, `IssuerType` in `{ACOR, CORP}`, `PrimaryExch` in `{N, A, Q}`, `ConditionalType='RW'`, and `TradingStatusFlg='A'`. Legacy CRSP share-code filters such as 10 and 11 are not available in the same way in CIZ, so we use the equivalent CIZ security-type and trading-status fields instead.

³In the baseline specification, a day is admissible only if `DlyRetMissFlg` is either absent or equal to NA. Thus, windows are excluded if they contain any day with `DlyRetMissFlg` in `{MV, NS, NT, RA, GP, MP, DG, DM, DP}`.

momentum and non-missing lagged market capitalization. Of the 60,910,165 rows in the expanded panel, 52,479,222 satisfy the momentum-validity requirement. After requiring non-missing lagged market capitalization, the final CRSP-based analysis sample contains 52,437,348 observations on 22,642 securities. The earliest date with valid momentum is January 2, 1980. Table IA.2 reports the corresponding attrition. At the stock-month level, 2,505,762 stock-months satisfy the momentum-validity requirement; the excluded observations consist of 304,805 stock-months with insufficient formation history, 63,184 stock-months containing disallowed return-status flags, and 47,116 stock-months containing trading-calendar-added empty observations.

7. **Momentum deciles and portfolio weights.** We assign each stock-month to a momentum decile using Kenneth French’s NYSE 2–12 prior-return breakpoints, aligned to the month immediately preceding the holding month. All stock-months that survive the post-momentum filters are successfully matched to a breakpoint date and receive a non-missing decile assignment. For the portfolio analysis, weights are computed within each (*date*, *decile*) cell using lagged market capitalization:

$$w_{i,t}^{(q)} = \frac{\text{MCAP}_{i,t-1}}{\sum_{j \in q(t)} \text{MCAP}_{j,t-1}},$$

where $q(t)$ denotes the set of securities in the relevant decile on date t , and $\text{MCAP}_{i,t-1}$ is security i ’s market capitalization on the previous trading day. These lagged market-cap weights are included in the final analysis panel.

8. **Additional daily characteristics.** We merge S&P 500 membership from the CRSP index constituents database. We also merge daily factors from Kenneth French’s data library (*Fama/French 3 Factors*), from which we use the daily risk-free rate (**RF**) and the daily market return (**Mkt**) in the analysis. In addition, we construct the bid-ask spread measure

$$\text{BAS}_{i,t} = \frac{\text{Ask}_{i,t} - \text{Bid}_{i,t}}{(\text{Ask}_{i,t} + \text{Bid}_{i,t})/2}.$$

The S&P 500 membership indicator is available for all observations; 5,440,227 daily rows correspond to index members and 46,997,121 correspond to non-members. The BAS variable is non-missing for 45,998,824 out of 52,437,348 observations, correspond-

ing to 87.72% coverage.

9. **TAQ sell-pressure measures.** We merge daily sell-pressure measures derived from WRDS TAQ intraday indicator data into the CRSP panel using PERMNO-date. The TAQ data are available from September 10, 2003 through October 27, 2022, so coverage is nonzero only within this period. The merge is performed as a left merge, so the number of daily panel rows is unchanged. Within the full 1980–2025 panel, 17,778,629 observations have non-missing net selling pressure and seller-initiated volume share, corresponding to 33.90% of the sample. Restricting to the TAQ-available period, the panel contains 18,105,868 observations on 9,407 securities, of which 98.19% have non-missing sell-pressure measures.
10. **Final panel.** The final dataset is a daily PERMNO-date panel spanning January 2, 1980 through December 31, 2025. It contains CRSP returns and market-cap variables, momentum measures and momentum deciles, the month-relative trading-day index, lagged market-cap portfolio weights, S&P 500 membership, daily factor variables, bid-ask spreads, and TAQ-based sell-pressure measures.

Table IA.1: CRSP Sample Construction

Step	Rows	Unique PERMNO
Deduplicated CRSP input	87,304,208	37,368
Common-stock filter	61,022,853	23,834
Exchange/status filter	59,917,630	23,827
Trading-day expansion	60,910,165	23,827

Table IA.2: Momentum Construction and Attrition

Step	Rows	Unique PERMNO
Input panel with month-relative T	60,910,165	23,827
After valid momentum requirement	52,479,222	22,644
After lagged market-cap filter	52,437,348	22,642

Mutual-fund sample. The filter we apply to identify the mutual-fund sample varies across specifications, because CRSP’s objective-code coverage and MFLINKS membership do not extend back to 1962 on a consistent basis. For the daily fund analysis (Sections 6.3 and 7), we restrict to actively managed U.S. equity diversified funds using a filtered identifier list constructed from MFLINKS membership, CRSP entity flags (`et_flag`), and index-fund exclusions, with an AUM threshold of \$10 million. For the monthly Carhart replication (Table 10, 1962–2025), we apply the AUM \$10 million filter to the CRSP fund universe without additional objective-code restriction, because consistent equity-fund classification fields are not available before the late 1970s. For analyses using MFLINKS-linked holdings or daily flow data (post-1998), we apply MFLINKS membership and active-fund flags directly. The substantive results in Table 10 are robust to applying the tighter MFLINKS-plus-flag filter on the post-1980 subsample: D1 monthly alpha differs by 0.91 bps/mo, D10 by 2.80 bps/mo, and UMD loadings differ by less than 0.01.

IA.2 Holding Period Analysis

A related question is whether the PreTOM concentration depends on the choice of holding period. Our baseline uses a one-month holding period ($K = 1$), which is standard in the factor zoo literature but shorter than the 3–12 month holding periods emphasized in Jegadeesh and Titman (1993). Table IA.3 addresses this by varying K from 1 to 12 months while keeping the formation period fixed at 12 months with a one-month skip, using NYSE-breakpoint decile assignments throughout. For $K > 1$, we implement the overlapping-portfolio construction of Jegadeesh and Titman (1993): the strategy holds K cohorts simultaneously, each formed in a different past month and each held for K months with fixed composition, and takes the equal-weighted average of their returns on each trading day. This is the standard approach for extending holding periods in the momentum literature (Carhart, 1997; Jegadeesh and Titman, 2001). Standard errors use Newey-West with $K - 1$ lags to account for the autocorrelation induced by the overlapping cohorts.

The PreTOM component is statistically significant at the 1% level for every holding period examined, while the Rest component is never significant. By $K = 9$, cumulative Rest wealth falls below 1 so PreTOM wealth exceeds total strategy wealth. By $K = 12$, the Rest component is negative, and the PreTOM share reaches 139%—meaning the six pre-month-end trading days generate more wealth than the full strategy, while all remaining trading days destroy it. The PreTOM concentration is not a feature of short holding periods. If anything, it becomes more pronounced as K increases.

This monotonicity is consistent with the month-end cash-demand mechanism. The information-driven component of momentum (which operates uniformly across trading days) decays quickly as the formation signal becomes stale, while the flow-driven PreTOM component (which reflects a monthly institutional cycle) persists as long as funds continue to hold stale losers.

Table IA.3: PreTOM Concentration by Holding Period

	$K = 1$	$K = 3$	$K = 6$	$K = 9$	$K = 12$
<i>Panel A: Loser – Market (bps/month)</i>					
PreTOM	−56.8*** (−4.44)	−55.9*** (−4.76)	−51.8*** (−4.63)	−48.4*** (−4.34)	−44.4*** (−4.22)
Rest	−34.0 (−1.52)	−31.7 (−1.50)	−24.3 (−1.27)	−16.7 (−0.92)	−9.5 (−0.54)
<i>Panel B: Winner – Market (bps/month)</i>					
PreTOM	3.5 (0.45)	0.7 (0.10)	2.3 (0.32)	0.6 (0.09)	−1.1 (−0.15)
Rest	−0.7 (−0.06)	−6.5 (−0.53)	−8.3 (−0.70)	−9.5 (−0.81)	−12.9 (−1.16)
<i>Panel C: WML (bps/month)</i>					
PreTOM	61.4*** (3.59)	58.2*** (3.85)	55.6*** (3.86)	50.4*** (3.60)	44.4*** (3.38)
Rest	38.4 (1.37)	29.8 (1.12)	19.5 (0.78)	10.6 (0.44)	−0.4 (−0.02)
<i>Panel D: Wealth (\$1 invested, 1980–2025)</i>					
\$PreTOM	18.78	16.91	15.81	12.43	9.33
\$Rest	2.37	1.79	1.25	0.86	0.53
\$Total	44.46	30.28	19.76	10.71	4.93
PreTOM share	77%	83%	93%	105%	139%

Notes. Momentum portfolios use the French 12-month formation period with a one-month skip and NYSE breakpoints for decile assignment. The holding period K varies from 1 to 12 months. For $K > 1$, the strategy holds K overlapping portfolios simultaneously and takes the equal-weighted average of their returns. Panels A–C report mean monthly compounded returns in basis points; t -statistics in parentheses use Newey–West standard errors with $K - 1$ lags. Market-adjusted returns subtract the raw market return. Each panel compounds its own daily series independently to monthly returns. Because compounding is non-linear, Panel C (WML) does not equal Panel B (Winner – Market) minus Panel A (Loser – Market): the difference is a Jensen / convexity term that reflects realized daily-return volatility differentials across the legs. Panel D reports cumulative wealth from \$1 invested over 1980–2025. PreTOM share = $\log(\$PreTOM) / \log(\$Total) \times 100$. Sample: 1980–2025.

IA.3 Transaction Cost Decomposition

Novy-Marx and Velikov (2016) show that momentum is among the most expensive anomalies to implement. We decompose the standard value-weighted momentum strategy into PreTOM and Rest components. Transaction costs are computed following Novy-Marx and Velikov (2016): each stock that changes momentum decile incurs a round-trip cost equal to one full quoted bid-ask spread (half-spread to liquidate the position plus half-spread to enter the replacement). Costs are value-weighted within each leg and summed over the long (winner) and short (loser) sides of the long-short.⁴

Table IA.4 reports the results. Over the full sample (1980–2025), gross WML averages +91.3 bps per month, of which +59.7 bps (65%) accrues during PreTOM. Monthly-rebalanced costs of 79.9 bps per month leave a net return of +11.5 bps. A PreTOM-only strategy that enters and exits each month faces a full round-trip cost of 317.1 bps, making it unviable in the full sample due to pre-decimalization spreads. Post-decimalization (2001–2025), quoted spreads shrink dramatically (losers 24.4 bps, winners 14.8 bps), reducing the full round-trip cost to 39.2 bps. The PreTOM-only strategy earns +54.1 bps gross and +14.9 bps net of costs. The Rest strategy is negative both gross (−26.9 bps) and net (−66.1 bps). In recent markets, all net return originates in the PreTOM window. The cost assigned to Rest-only is conservative. Rest-only remains invested across the calendar-month boundary and therefore incurs an additional partial rebalance each month, since decile assignments change between months. Incorporating this additional turnover would further reduce Rest-only’s net return relative to PreTOM-only; the asymmetry we document is therefore understated by the symmetric round-trip treatment.

⁴Our cost measure is conservative. Novy-Marx and Velikov (2016) use the Hasbrouck (2009) effective spread, which is typically smaller than the quoted spread. Monthly turnover averages 20.9% for losers and 26.2% for winners.

Table IA.4: Transaction Cost Decomposition of Monthly Momentum Returns

	Gross (bps/mo)	TC (bps/mo)	Net (bps/mo)
<i>Panel A: Full sample (1980–2025)</i>			
Buy-and-hold WML	+91.3	79.9	+11.5 (0.35)
PreTOM component	+59.7*** (3.60)		
Rest component	+32.4 (1.12)		
PreTOM-only strategy	+59.7	317.1	−257.5 (−11.34)
Rest-only strategy	+32.4	317.1	−284.8 (−8.99)
<i>Panel B: Post-decimalization (2001–2025)</i>			
Buy-and-hold WML	+25.1	10.1	+15.0 (0.29)
PreTOM component	+54.1** (2.07)		
Rest component	−26.9 (−0.59)		
PreTOM-only strategy	+54.1	39.2	+14.9 (0.56)
Rest-only strategy	−26.9	39.2	−66.1 (−1.43)

Notes: The table decomposes monthly returns of the standard value-weighted WML strategy (long decile 10, short decile 1, rebalanced monthly using fixed momentum decile breakpoints from CRSP) into PreTOM ($[T-9, T-4]$) and rest-of-month components. Daily value-weighted returns within each window are compounded to monthly frequency. Transaction costs are computed following Novy-Marx and Velikov (2016) using quoted bid-ask spreads. “Buy-and-hold” charges one full quoted spread for each stock that changes momentum decile (monthly turnover: losers 21.0%, winners 26.3%). “PreTOM-only” and “Rest-only” strategies additionally incur a full round-trip spread on all portfolio stocks each month, since the investor enters and exits the entire long-short position within the month; Rest-only further bears a partial decile-rebalancing cost at each month boundary, which is not separately accounted for in the table. Post-decimalization, mean quoted spreads are 24.4 bps (losers) and 14.8 bps (winners), making the PreTOM-only strategy viable net of costs. t -statistics in parentheses, placed beneath the number being tested. ** $p < 0.05$, *** $p < 0.01$.

IA.4 T+1 Settlement Falsification Tests

Three checks confirm that the T+1 settlement result is specific to the boundary days the mechanism predicts and is not the artifact of a slow pre-existing drift. First, *placebo days*: running the identical $T-4$ vs. $T-3$ DiD but shifted to $T-6$ vs. $T-7$, two days well inside both the T+2 and T+1 selling windows where no boundary shifted, produces a DiD of -2.3 ($t = -0.08$). Second, *placebo dates*: the same $T-4$ vs. $T-3$ comparison using fake event dates spread across 1999–2015 (May 28 of 1999, 2005, 2011, 2013, and 2015), each well removed from major market disruptions, returns DiDs ranging from -13.4 to $+3.5$ with $|t| \leq 1.19$. All falsifications are indistinguishable from zero, in contrast to the $+84.7$ ($t = 2.80$) for the actual May 2024 reform. Table IA.5 reports the full results. Third, the *event study* below shows that the within-month $T-4$ minus $T-3$ differential has no pre-trend in the years before the reform and lifts only after May 2024, supporting the parallel-trends assumption underlying the DiD.

Table IA.5: T+1 Settlement: Falsification Tests

Test	DiD	t -stat	p -value	Prediction
Real: Losers, $T-4$ vs. $T-3$, May 2024	84.72	2.80	0.005	Nonzero
<i>Placebo days (losers, May 2024):</i>				
$T-6$ vs. $T-7$	-2.33	-0.08	0.936	Zero
<i>Placebo dates (losers, $T-4$ vs. $T-3$):</i>				
May 28, 2015	3.54	0.19	0.847	Zero
May 28, 2013	3.07	0.19	0.850	Zero
May 28, 2011	3.01	0.20	0.838	Zero
May 28, 2005	-10.05	-0.79	0.432	Zero
May 28, 1999	-13.41	-1.19	0.233	Zero

Notes: Falsification tests for the T+1 settlement DiD reported in the main paper. All tests use the same specification: market-adjusted VW loser returns regressed on a day indicator, a post-event indicator, and their interaction, with robust standard errors. Placebo days shift the comparison to $T-6$ vs. $T-7$, two days interior to both the T+2 and T+1 selling windows where no shift is predicted. Placebo dates use fake event dates when no regulatory change occurred. All falsification interactions are small and statistically insignificant. Sample: 1980–2025.

Event-study pre-trend test. The pooled DiD restricts the data to a binary pre/post split at May 28, 2024. A pre-trend in the within-month $T-4$ minus $T-3$ differential before that date would invalidate the DiD even if the placebo tests above were clean. We test this directly by replacing the binary Post_t indicator with a vector of 12-month event-year bins anchored at May. Bin 0 covers May 2024–April 2025, bin -1 (the omitted reference) covers May 2023–April 2024, bin -2 covers May 2022–April 2023, and so on through bin -8 . All months from 1980 through April 2015 are pooled into a single “deep pre” bin. May 2024 is excluded throughout, matching the main-text spec. On the subsample $[T-4, T-3]$, we estimate

$$r_t = \alpha + \sum_k \alpha_k \mathbf{1}[\text{bin}_t = k] + \delta \mathbf{1}[T = -4] + \sum_{k \neq -1} \beta_k \mathbf{1}[\text{bin}_t = k] \times \mathbf{1}[T = -4] + \varepsilon_t,$$

where r_t is the daily market-adjusted value-weighted loser return in basis points. Each β_k is the within-month $T-4$ minus $T-3$ differential in event-year bin k , relative to the year just before the reform.

Figure IA.1 plots the eleven coefficients with 95% confidence intervals. The eight pre-event coefficients are scattered around zero, none significant at the 5% level and with no monotone drift toward May 2024; the joint test that all eight equal zero cannot be rejected ($F(8, 1080) = 0.59$, $p = 0.78$). Both post-event bins lift to similar positive magnitudes: +121.3 basis points for May 2024–April 2025 ($t = 2.00$, $p = 0.05$) and +111.1 for the partial bin May 2025–December 2025 ($t = 1.81$, $p = 0.07$), each on the order of the pooled DiD reported in the main paper. The flat pre-period, the discontinuous post-period jump, and the persistence of the effect into the second post-event bin together support the parallel-trends assumption.

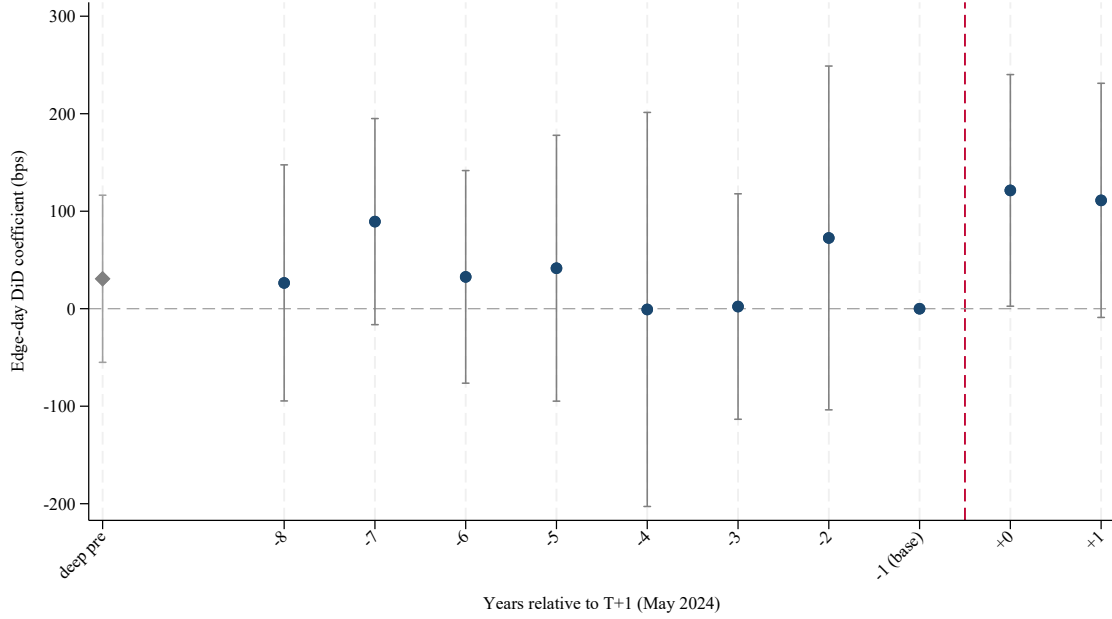
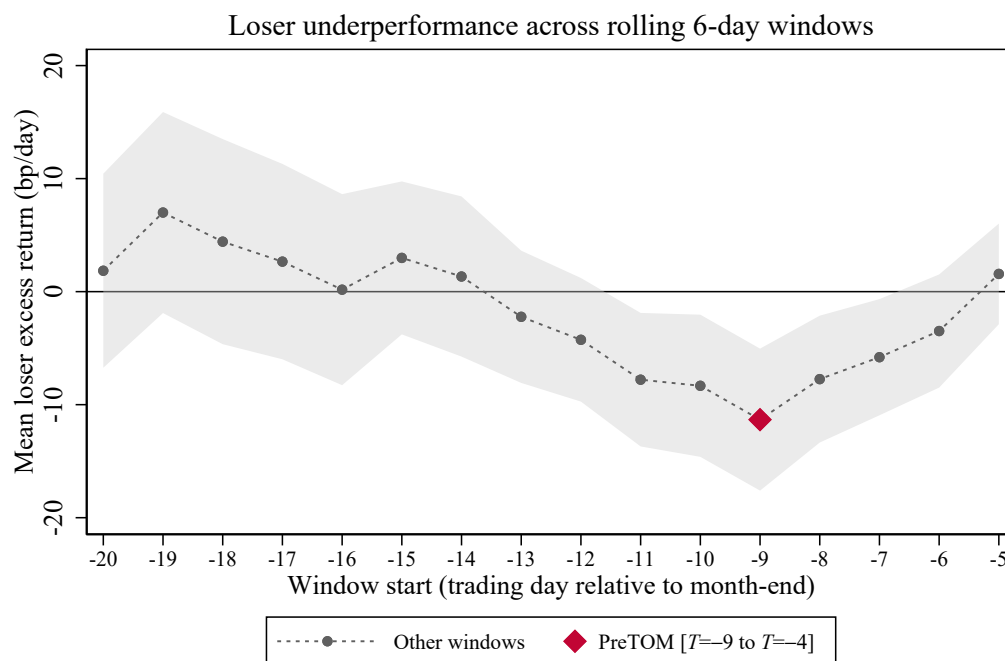


Figure IA.1: Event study of the $T-4$ vs. $T-3$ DiD. Each point is the within-month $T-4$ minus $T-3$ differential in market-adjusted value-weighted loser returns (basis points), estimated for the indicated 12-month bin relative to the omitted bin -1 (May 2023–April 2024). Bins are anchored at May, so bin 0 is May 2024–April 2025 and bin -2 is May 2022–April 2023. The deep-pre bin pools 1980 through April 2015. Whiskers are 95% confidence intervals from heteroskedasticity-robust standard errors. May 2024 is excluded. The joint F -test of all eight pre-event coefficients equal to zero gives $F(8, 1080) = 0.59$, $p = 0.78$. Sample: 1980–2025.

IA.5 Sliding Window Analysis

A natural concern is that the PreTOM window $[T-9, T-4]$ was selected ex post to maximize the effect. We address this by sliding a six-day window across the second half of the trading month and computing the raw mean VW loser excess return ($\text{loser} - r_f$) within each window. We restrict to windows starting on or after trading day 7 (approximately mid-month) to ensure that all days within each window belong to the same monthly portfolio assignment.

The mean loser excess return is near zero for mid-month windows, declines monotonically as the window shifts toward month-end, and troughs at PreTOM ($[T-9, T-4]$): -11.32 basis points per day ($t = -3.53$). No other window starting position produces a comparably negative estimate. The decline into PreTOM and partial recovery at month-end are exactly what the month-end cash-demand mechanism predicts.



Mean VW loser-market return per 6-day window. Only months with all 6 days in the window included. Newey-West SEs (5 lags) on monthly means. 1980-2025.

Figure IA.2: Sliding six-day window analysis. Each point plots the mean daily VW loser excess return ($\text{loser} - r_f$) for a six-day window starting at the indicated trading day. The trough occurs at PreTOM ($[T-9, T-4]$). Sample: 1980–2025.

IA.6 Tail Trimming Analysis

A key concern is whether PreTOM profitability is structural or driven by a few extreme days. We address this by progressively trimming the worst days from each window's return distribution and recomputing the mean. Figure IA.3 shows the result. PreTOM mean WML is +10.2 bps untrimmed and rises to +17.2 bps after removing the worst 1% of days, confirming that the effect is structural and not driven by outliers. In contrast, month-start mean WML flips from -4.4 bps to +6.2 bps after the same 1% trim: approximately 15 extreme days drive the entire negative month-start average. This asymmetry confirms that PreTOM profits and month-start crashes have distinct statistical signatures: PreTOM is a persistent daily effect, while month-start negativity is concentrated in rare tail events.

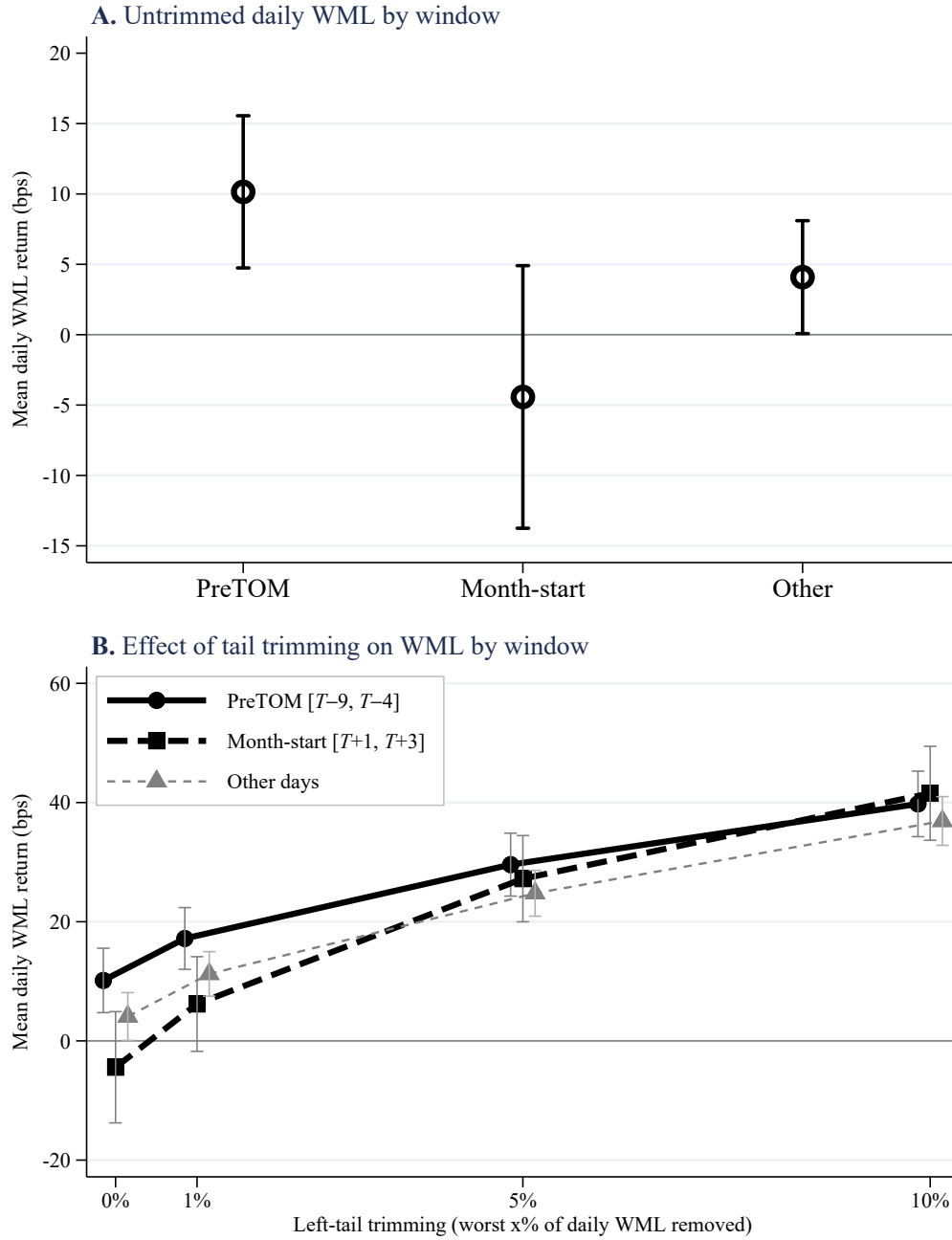


Figure IA.3: Mean daily WML return by calendar window after progressively trimming the worst days from each window's distribution. Panel A: pre-month-end ($[T-9, T-4]$). Panel B: month-start ($[T+1, T+3]$). PreTOM mean WML rises from +10.2 to +17.2 bps after 1% trimming (structural), while month-start flips from -4.4 to +6.2 bps (driven entirely by ~ 15 tail events). Sample: 1980–2025, value-weighted.

IA.7 Crash-Day Concentration: Calendar-Share Methodology

This appendix details the calendar-share construction and proportions z -test summarized in Section 8 of the main paper.

Crash-day classification. We classify a trading day as a crash day if the value-weighted WML return falls below -200 basis points. Robustness at -100 and -300 basis-point thresholds is reported in the figures.

Comparison universes. To prevent the PreTOM and month-start windows from contaminating each other, we compare each window’s crash share against its calendar share within a comparison universe that excludes the other window. For PreTOM, the universe is non-month-start trading days (PreTOM days plus other days, excluding $[T+1, T+3]$). For month-start, the universe is non-PreTOM trading days (month-start days plus other days, excluding $[T-9, T-4]$). The calendar share is the fraction of trading days in the universe that fall in the window of interest. Under the null that crashes are spread proportionally across the calendar, the observed share equals the calendar share.

Test statistic. For each window w and threshold, let $p_{\text{obs}} = n_{\text{crash},w}/n_{\text{crash}}$ be the observed share of crash days in window w and p_{exp} be the calendar share. The standard one-sample proportions z -statistic is

$$z = \frac{p_{\text{obs}} - p_{\text{exp}}}{\sqrt{p_{\text{exp}}(1 - p_{\text{exp}})/n_{\text{crash}}}},$$

two-sided p -values from the standard normal.

Excluded episodes. The robustness reported in the body excludes 11 months that contain famous momentum-crash episodes: January 2001, October–November 2001, and November 2002 (the Daniel–Moskowitz 2016 cluster); March–May 2009; and February–May 2020 (COVID dislocation).

IA.8 International Evidence: Methodology

This appendix details the construction of the international momentum decile portfolios used in the international evidence presented in the main paper.

Sample. Daily stock returns are from Compustat Global (`comp.g.secd`) for 19 developed markets: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Hong Kong, Ireland, Italy, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, and the United Kingdom. Country is identified by the issuer-of-record (`loc`) field rather than the listing exchange. The sample begins in 1990 for most countries and in 1993 for Hong Kong and Switzerland, where Compustat coverage is thin in the early 1990s. We exclude Japan, where momentum returns are approximately zero over our sample period (Fama and French, 2012; Asness et al., 2013), and Canada, for which Compustat Global coverage is limited (Canadian common stocks are concentrated in the CRSP North-America universe).

Stock universe and cleaning. We retain common stocks only (`tpci='0'`). The total daily return is constructed from price (`prccd`), the cumulative split-adjustment factor (`ajexdi`), and the cumulative dividend reinvestment factor (`trfd`):

$$r_{i,t} = \frac{p_{i,t}/a_{i,t} \cdot \tau_{i,t}}{p_{i,t-1}/a_{i,t-1} \cdot \tau_{i,t-1}} - 1.$$

We drop observations with missing return or market capitalization, non-positive market capitalization, and returns outside $[-0.99, 10]$ to remove obvious data errors. Market capitalization is `prccd` $\times$ `cshoc`.

Monthly aggregation. Daily returns are compounded to monthly returns within each stock-month. A stock-month is included only if it contains at least 15 daily observations; this rules out partially-traded months that would otherwise distort the momentum signal.

Momentum signal. For each formation month m , the signal is the cumulative log return over the eleven months $m-12$ through $m-2$, requiring all eleven monthly returns to be observed. Deciles are formed at the end of m and held in calendar month $m+1$. This follows

the standard 12-2 convention used in Asness et al. (2013) and the broader international momentum literature.

Decile assignments. Within each country-month, we rank stocks into momentum deciles using *size-conditional* breakpoints: decile cutoffs are computed from stocks above the within-country median market capitalization at the prior month-end, then all stocks (including small-cap) are assigned to deciles based on those cutoffs. This avoids decile assignments being driven by micro-cap noise. For seven countries with thin large-cap coverage we pool breakpoints within a broader region: small-Europe (Austria, Belgium, Finland, Ireland, Portugal) and small-Asia-Pacific (New Zealand, Singapore). All other countries (Australia, Denmark, France, Germany, Hong Kong, Italy, the Netherlands, Norway, Spain, Sweden, Switzerland, the United Kingdom) compute breakpoints from their own large-cap stocks each month. Decile assignments are fixed within each calendar month.

Returns and market adjustment. Value-weighted portfolio returns use prior-month-end market capitalization as weights. The regional market excess return is taken from Kenneth French’s regional factor files: *Europe 3 Factors Daily* for the European countries, *Asia Pacific ex Japan 3 Factors Daily* for Australia, Hong Kong, New Zealand, and Singapore. We subtract the relevant regional market excess return to obtain market-adjusted returns. The dependent variable in equation (10) is the day- t value-weighted loser-decile return in country c in excess of its regional market return, scaled to basis points.

Code reference. The full pipeline is implemented in `_intl_build_v3.py` in the replication archive. The output of this pipeline is the country-day panel `compustat_intl_pooled_v3.csv` on which Table 9 regressions are estimated.

Full country tables. Tables IA.6 and IA.7 report PreTOM, Rest, and Difference estimates for all 19 countries individually, for both legs (Losers – Market and Winners – Market). Countries are sorted by the loser-leg Difference column.

Table IA.6: International Evidence: All Countries — Losers – Market (bps/day)

Country	PreTOM		Rest of Month		Difference	
	Mean	<i>t</i>	Mean	<i>t</i>	Mean	<i>t</i>
Ireland	-7.36	(-1.34)	4.95	(1.17)	-12.32*	(-1.80)
Norway	-12.10***	(-3.26)	-0.73	(-0.32)	-11.37***	(-2.63)
Netherlands	-7.93*	(-1.96)	2.65	(1.14)	-10.57**	(-2.31)
Hong Kong	-9.15***	(-2.73)	-0.21	(-0.09)	-8.94**	(-2.21)
Sweden	-4.62	(-1.19)	3.68*	(1.71)	-8.30*	(-1.90)
Finland	-5.98	(-1.57)	1.92	(0.65)	-7.90	(-1.61)
Australia	-8.23***	(-3.28)	-1.56	(-0.84)	-6.67**	(-2.24)
Switzerland	-4.59	(-0.98)	1.86	(0.78)	-6.45	(-1.19)
Belgium	-6.55**	(-2.08)	-1.78	(-0.61)	-4.77	(-1.11)
France	-4.85*	(-1.77)	-0.18	(-0.08)	-4.67	(-1.35)
Spain	-3.80	(-1.19)	0.35	(0.21)	-4.15	(-1.14)
New Zealand	-2.91	(-0.67)	0.87	(0.30)	-3.78	(-0.66)
Denmark	-3.79	(-1.18)	-1.09	(-0.51)	-2.70	(-0.68)
Germany	-5.05*	(-1.81)	-2.75	(-1.26)	-2.30	(-0.63)
UK	-3.09	(-1.06)	-1.35	(-0.65)	-1.74	(-0.45)
Portugal	-3.22	(-0.84)	-3.41	(-1.29)	0.19	(0.04)
Italy	-2.78	(-0.91)	-5.65*	(-1.93)	2.87	(0.69)
Singapore	6.16**	(2.09)	2.69	(1.33)	3.47	(1.01)
Austria	-0.57	(-0.17)	-5.63**	(-2.14)	5.06	(1.21)
<i>Pooled</i>	-4.76***	(-5.82)	-0.28	(-0.49)	-4.48***	(-4.46)

Notes: Daily VW portfolio returns from Compustat Global, 1990–2025 (HKG/CHE from 1993). Momentum signal is the cumulative log return over months -12 through -2 , requiring all eleven months observed (Asness et al., 2013 convention). NYSE-style breakpoints (above-median mcap). Countries with thin large-cap coverage use regional breakpoints (small-Europe: AUT, BEL, FIN, IRL, PRT; small-Asia-Pacific: NZL, SGP). Market-adjusted using Kenneth French regional factors. Sorted by Difference. Individual *t*-statistics clustered by month; pooled includes country FE in the Difference column and clusters by country-month.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.7: International Evidence: All Countries — Winners — Market (bps/day)

Country	PreTOM		Rest of Month		Difference	
	Mean	<i>t</i>	Mean	<i>t</i>	Mean	<i>t</i>
Ireland	1.26	(0.28)	1.31	(0.44)	−0.05	(−0.01)
Norway	2.37	(0.63)	5.58***	(2.65)	−3.21	(−0.74)
Netherlands	5.89	(1.46)	1.43	(0.70)	4.46	(0.98)
Hong Kong	−3.21	(−0.86)	1.26	(0.59)	−4.47	(−1.01)
Sweden	5.06*	(1.73)	4.51***	(2.70)	0.55	(0.15)
Finland	3.19	(0.95)	2.76	(1.18)	0.43	(0.10)
Australia	5.99**	(2.27)	4.83***	(2.82)	1.16	(0.35)
Switzerland	5.95**	(2.17)	1.58	(0.85)	4.37	(1.27)
Belgium	−5.79*	(−1.84)	3.71	(1.47)	−9.50**	(−2.38)
France	1.92	(0.81)	2.15	(1.18)	−0.23	(−0.07)
Spain	5.51*	(1.94)	3.65**	(2.05)	1.87	(0.55)
New Zealand	4.21	(0.93)	1.81	(0.73)	2.39	(0.46)
Denmark	5.15	(1.54)	4.47**	(2.22)	0.68	(0.16)
Germany	6.34*	(1.65)	2.53	(1.29)	3.81	(1.02)
UK	0.94	(0.31)	2.73	(1.63)	−1.79	(−0.50)
Portugal	6.68	(1.38)	4.75*	(1.86)	1.93	(0.35)
Italy	4.55*	(1.86)	3.00*	(1.75)	1.55	(0.52)
Singapore	4.08	(1.16)	1.98	(0.97)	2.10	(0.49)
Austria	5.37	(1.63)	1.42	(0.52)	3.94	(0.93)
<i>Pooled</i>	3.44***	(4.35)	2.92***	(6.01)	0.52	(0.54)

Notes: Same specification as Table IA.6. Countries sorted in the same order as the losers table for comparability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.9 Subperiod PreTOM Concentration: VW versus EW

Table IA.8 reports daily PreTOM and Rest mean returns for D1 (loser–Mkt) and WML by historical subperiod under both value-weighting (Panel A) and equal-weighting (Panel B). PreTOM days are 29% of the trading month. If daily returns were generated uniformly across the month, PreTOM and Rest mean returns would coincide up to sampling error; any systematic gap between them reflects a calendar-time concentration in the underlying return process.

PreTOM losses dominate Rest-of-month returns in every subperiod from 1927 onward, but the gap widens sharply after 1962, coinciding with the scaling of institutional ownership in U.S. equities.

Table IA.8: Subperiod Analysis: PreTOM versus Rest, D1 (Loser – Mkt) and WML

<i>Panel A: D1 (Loser – Mkt) returns (bps/day)</i>					
Period	Value-weighted		Equal-weighted		Inst. Own.
	PreTOM	Rest	PreTOM	Rest	
1927–1940	–5.43 (–1.07)	–4.86 (–1.74)	19.31 (3.50)	23.13 (7.34)	< 10%
1940–1962	–2.75 (–2.08)	–2.45 (–2.78)	1.81 (1.19)	3.96 (3.62)	< 10%
1962–1980	–10.09*** (–6.90)	–4.63 (–4.74)	–3.74*** (–2.58)	2.53 (2.39)	15–30%
1980–2002	–12.18* (–5.96)	–6.68 (–5.09)	5.92 (2.84)	8.85 (6.60)	30–55%
2002–2025	–7.12** (–2.04)	1.50 (0.65)	2.42 (0.95)	3.00 (1.71)	55–72%
<i>Panel B: WML (Winner – Loser) returns (bps/day)</i>					
Period	Value-weighted		Equal-weighted		Inst. Own.
	PreTOM	Rest	PreTOM	Rest	
1927–1940	2.25 (0.34)	9.10 (2.32)	–13.59 (–2.38)	–12.06 (–3.63)	< 10%
1940–1962	5.75 (2.92)	4.85 (3.75)	3.58 (1.88)	1.10 (0.86)	< 10%
1962–1980	10.74 (4.91)	7.92 (5.56)	6.80* (4.29)	2.40 (2.10)	15–30%
1980–2002	10.24 (3.40)	7.06 (3.77)	–6.01 (–3.28)	–4.66 (–3.81)	30–55%
2002–2025	10.06** (2.21)	–1.91 (–0.64)	1.08 (0.42)	–3.12 (–1.73)	55–72%

Notes. Daily mean returns in basis points per day, by historical subperiod and weighting scheme. Panel A: D1 (loser – Mkt) portfolio return. Panel B: WML (winner – loser) excess return. PreTOM is the trading-day window $[T-9, T-4]$ before month-end; Rest is all other trading days. PreTOM and Rest level cells report mean and robust t -statistic; the diff t -statistic that drives the stars uses Newey-West (21 lags) on the joint regression of the daily series on PreTOM. *Asterisks on PreTOM cells denote significance of the PreTOM versus Rest difference*, not significance of the level. Aggregate institutional ownership figures are approximate ranges from Blume and Keim (2012) and the NYSE Fact Book. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.10 Selling Recency: Fresh Losers Bear Disproportionate PreTOM Pressure

This section reports the detail behind the fresh-versus-stale loser result summarized in Section 4.2.3 of the main paper. Within the bottom momentum decile, we measure “freshness” as the cumulative return over the most recent three months ($\text{ret}_{3,1}$), standardized cross-sectionally within D1 each month and sign-flipped so that higher values indicate worse recent performance. All stocks in D1 share similar 12-month losses (the sorting variable), so freshness captures the *timing* of the loss, not its magnitude: fresh losers (bottom tercile, mean $\text{ret}_{3,1} = -33.8\%$) and stale losers (top tercile, mean $\text{ret}_{3,1} = +14.9\%$) have nearly identical 12-month returns (-56.3% vs. -45.2%).

Table IA.9 reports the regressions. Panel A splits D1 into terciles; fresh losers underperform stale losers by 4.5 bps per day during PreTOM ($F = 7.15$, $p = 0.008$). Panel B uses a continuous freshness measure: the PreTOM $\times$ freshness effect is -1.5 bps per standard deviation without controls ($t = -2.10$), strengthening to -2.2 ($t = -2.76$) after controlling for the prior-month bid-ask spread $\text{BAS}_{i,m-1}$. The effect vanishes in December: the December $\times$ PreTOM $\times$ freshness interaction is $+3.5$ ($t = 1.44$), leaving the total December effect at $+1.1$ ($t = 0.44$). Outside December the effect strengthens to -2.5 ($t = -3.04$), consistent with tax-loss harvesting depleting the fresh-loser pool before PreTOM begins.

Table IA.9: PreTOM Returns by Loser Freshness

	Coefficient	t -statistic	p -value
<i>Panel A: Tercile split within D1 losers</i>			
PreTOM $\times$ Fresh (bottom tercile $\text{ret}_{3,1}$)	-3.30^{**}	(-2.17)	0.030
PreTOM $\times$ Stale (top tercile $\text{ret}_{3,1}$)	$+1.24$	(1.02)	0.308
Test Fresh = Stale	$F = 7.15, \quad p = 0.008$		
<i>Panel B: Continuous freshness measure</i>			
Without BAS controls ($N = 9,954,220$)			
PreTOM $\times$ fresh_z	-1.51^{**}	(-2.10)	0.036
With $\text{BAS}_{i,m-1}$ controls ($N = 7,093,974$)			
PreTOM $\times$ fresh_z	-2.19^{***}	(-2.76)	0.006
PreTOM $\times$ $\text{BAS}_{i,m-1}$	$+36.59^{***}$	(3.45)	0.001
<i>Panel C: December falsification (with $\text{BAS}_{i,m-1}$ controls)</i>			
PreTOM $\times$ fresh_z (non-December)	-2.47^{***}	(-3.04)	0.002
December $\times$ PreTOM $\times$ fresh_z	$+3.54$	(1.44)	0.149
Total December effect (sum)	$+1.06$	(0.44)	0.658

Notes: Sample consists of bottom-decile (D1) momentum losers only, 1980–2025. Freshness is measured by $\text{ret}_{3,1}$ (cumulative return over months -3 through -1), standardized within D1 each month and sign-flipped so that higher values indicate worse recent performance. Panel A splits D1 into terciles by freshness; PreTOM $\times$ Fresh and PreTOM $\times$ Stale are the window effects for the bottom and top terciles, and the F -test examines their equality. Panel B uses the continuous standardized measure, with and without bid-ask spread controls. Panel C interacts the continuous measure with a December indicator and reports the total December effect (non-December coefficient plus the December interaction). $\text{BAS}_{i,m-1}$ is the mean daily bid-ask spread of stock i over the prior calendar month $m-1$, as a fraction of the quote midpoint; stock-months with fewer than 10 valid prior-month BAS days are excluded. All specifications include firm and date fixed effects; standard errors are double-clustered by firm and date. Returns in basis points per day. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. $^{**}p < 0.05$, $^{***}p < 0.01$.

IA.11 Window Dressing Falsification

Brown (2017) documents that momentum returns concentrate in the third month of calendar quarters, attributing the pattern to tax-loss selling and window dressing by delegated managers.

If window dressing is the primary driver of the pre-month-end loser effect, the pattern should amplify in December and at quarter-ends. Our evidence points in the opposite direction. The PreTOM effect is present every month, with no quarter-end amplification, and is not absorbed by December tax-driven selling.

Two tests address the window dressing alternative directly, both reported in Table IA.10. First, restricting the sample to non-quarter-end months, where window dressing is least relevant, the Loser $\times$ PreTOM coefficient is -7.5 bps ($t = -2.66$), virtually identical to the full-sample estimate. Second, interacting Loser $\times$ PreTOM with a quarter-end indicator produces a small, insignificant triple interaction: there is no systematic quarter-end amplification, consistent with a monthly liquidity cycle rather than quarterly reporting incentives.

A parallel test isolates December, the month in which tax-loss harvesting is most concentrated. Restricting the sample to non-December months yields a Loser $\times$ PreTOM coefficient of -7.03 bps ($t = -2.86$, $N = 48.85$ million, untabulated), virtually identical to the full-sample estimate of -7.15 . The PreTOM concentration is not driven by December tax-loss selling.

Table IA.10: Window Dressing Falsification

	(1) Non-QE months	(2) Full sample, QE interaction	(3) Full sample baseline
Loser	4.132*** (2.68)	4.039*** (2.62)	2.609** (2.04)
Loser $\times$ PreTOM	-7.523*** (-2.66)	-7.496*** (-2.65)	-7.151*** (-3.08)
Loser $\times$ QtrEnd		-4.249 (-1.48)	
Loser $\times$ PreTOM $\times$ QtrEnd		0.990 (0.18)	
Constant	4.636*** (33.84)	3.442*** (29.93)	3.442*** (29.42)
Fixed effects	Firm, Date	Firm, Date	Firm, Date
Observations	35,417,562	53,324,145	53,324,145
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable is the daily market-adjusted return ($r_{i,t} - r_t^m$, in excess of the value-weighted CRSP market return), in basis points. All specifications are value-weighted using lagged market-capitalization weights. Column (1) restricts the sample to non-quarter-end months (Jan, Feb, Apr, May, Jul, Aug, Oct, Nov); window dressing would be least relevant in these months, yet the Loser $\times$ PreTOM coefficient is essentially identical to the full-sample estimate. Column (2) estimates the full sample with quarter-end interactions; QtrEnd equals one in March, June, September, December. The triple interaction Loser $\times$ PreTOM $\times$ QtrEnd is small and statistically indistinguishable from zero, indicating no quarter-end amplification. Column (3) reproduces the full-sample baseline from Table 2 of the main paper for reference. All specifications include firm and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by firm and date. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: 1980–2025.

IA.12 Dividend Cycle: Within-Quarter Detail

This section reports the detail behind the dividend-cycle pattern summarized in Section 4.2.2 of the main paper. A dividend payment delivers a cash buffer that drains over the months until the next payment. We examine whether PreTOM selling pressure on a stock varies with its position in the payment cycle, and whether any such pattern is concentrated in losers rather than winners. We do not derive the within-quarter pattern as a strict implication of the dispensability mechanism, since stock-specific dividend timing need not translate into stock-specific selling pressure when institutions manage cash at the portfolio level. We present the within-cycle estimates as descriptive evidence consistent with a within-cycle modulation of cash needs at the stock level.

We test this within D1 and D10 quarterly payers. With $\text{Month}_{i,t} \in \{1, 2, 3\}$ indexing the position of day t within stock i 's dividend quarter ($m = 1$: dividend month; $m = 2$: one month after; $m = 3$: two months after), we estimate

$$\begin{aligned} r_{i,t} - r_t^m = & \alpha \text{Loser}_{i,t} + \sum_{m=1}^3 \beta_m^L \text{Loser}_{i,t} \times \text{PreTOM}_t \times \mathbf{1}[\text{Month}_{i,t} = m] \\ & + \sum_{m=1}^2 \beta_m^W \text{PreTOM}_t \times \mathbf{1}[\text{Month}_{i,t} = m] + \sum_{m=1}^2 \lambda_m \text{Loser}_{i,t} \times \mathbf{1}[\text{Month}_{i,t} = m] \\ & + \sum_{m=1}^2 \theta_m \mathbf{1}[\text{Month}_{i,t} = m] + \mu_i + \delta_t + \varepsilon_{i,t}. \end{aligned}$$

Month 3 is the omitted reference for the lower-order Month and Loser $\times$ Month interactions. Standard errors are two-way clustered by firm and date. The standalone $\text{Loser}_{i,t}$ captures the within-firm time-varying component of decile membership; firm fixed effects μ_i absorb only its firm-level mean. Date fixed effects δ_t absorb the standalone PreTOM_t indicator. Table IA.11 reports the estimates in Panel B (returns) and Panel C (TAQ).⁵

Both empirical patterns appear in the TAQ data. Panel C of Table IA.11 reports the loser coefficients rising monotonically across the cycle: $\gamma_1^L = +0.0018$ ($t = 0.70$), $\gamma_2^L = +0.0051$

⁵Coefficients β_m^L report the loser-specific Loser $\times$ PreTOM $\times$ Month effect in each dividend cycle month rather than as differentials from Month 3; the saturated triple-interaction structure identifies all three m jointly with the $\alpha \text{Loser}_{i,t}$ standalone. Winner coefficients β_m^W for $m \in \{1, 2\}$ are PreTOM $\times$ Month effects for non-losers relative to the Month 3 reference. The TAQ analog replaces $r_{i,t} - r_t^m$ with $\text{NSP}_{i,t}$ and relabels $\beta \rightarrow \gamma$; the right-hand side is identical.

($t = 1.84$), $\gamma_3^L = +0.0057$ ($t = 2.12$). The winner coefficients are statistically flat: $\gamma_1^W = +0.0001$ ($t = 0.06$), $\gamma_2^W = +0.0016$ ($t = 0.77$); the joint test $\gamma_1^W = \gamma_2^W = 0$ does not reject ($F = 0.38$, $p = 0.68$). The dividend-cycle pattern is loser-specific in the data.

Panel B (returns) is directionally consistent but underpowered to detect cross-month differences. Loser coefficients are $\beta_1^L = -6.3$ bps/day ($t = -1.32$), $\beta_2^L = -10.6$ ($t = -1.98$), $\beta_3^L = -7.7$ ($t = -1.57$); the joint Wald does not reject equality across months ($F = 0.31$, $p = 0.73$). Winner coefficients are flat ($\beta_1^W = -4.4$, $\beta_2^W = -0.9$). Trade-direction is more informative than returns at this sample size, consistent with the broader pattern that institutional selling shows more directly in TAQ flows than in daily returns.

Table IA.11: Dividend Income and PreTOM Selling Pressure

<i>Panel A: Dividend Payer Composition</i>			
	D1 Losers	D10 Winners	
Non-payer	84.0%	67.6%	
Quarterly payer	8.8%	23.2%	
Other (semi-annual, annual, monthly)	7.2%	9.2%	
<i>Panel B: Within-Quarter Returns (Quarterly Payers, D1 $\cup$ D10)</i>			
	Month 1 (just paid)	Month 2 (1 mo after)	Month 3 (about to pay)
Winners: PreTOM $\times$ Month _m	−4.35 (−1.48)	−0.87 (−0.28)	0 (ref.)
Losers: Loser $\times$ PreTOM $\times$ Month _m	−6.28 (−1.32)	−10.56** (−1.98)	−7.74 (−1.57)
<i>N</i>		2,449,358	
<i>Panel C: Within-Quarter TAQ Net Selling Pressure (Quarterly Payers, D1 $\cup$ D10)</i>			
	Month 1 (just paid)	Month 2 (1 mo after)	Month 3 (about to pay)
Winners: PreTOM $\times$ Month _m	+0.0001 (0.06)	+0.0016 (0.77)	0 (ref.)
Losers: Loser $\times$ PreTOM $\times$ Month _m	+0.0018 (0.70)	+0.0051* (1.84)	+0.0057** (2.12)
<i>N</i>		768,018	

Notes: Panel A reports the share of stock-month observations by dividend frequency classification (CRSP distributions, modal frequency). Panels B and C restrict to quarterly dividend payers within D1 (loser decile) and D10 (winner decile) momentum portfolios. Month 1 = the dividend-payment month; Month 2 = one month after payment; Month 3 = about to pay again. Both panels report estimates from the triple-interaction specification described in Online Appendix Section IA.12; Panel C uses the same right-hand side with NSP_{*i,t*} replacing ExRet_{*i,t*} on the left. The “Winners” row reports β_m^W , the PreTOM $\times$ Month_m coefficient identified from D10 (winners are the omitted-loser baseline) for $m \in \{1, 2\}$ relative to the omitted reference Month 3. The “Losers” row reports β_m^L , the Loser $\times$ PreTOM $\times$ Month_m triple-interaction coefficient: these three coefficients are saturated across $m \in \{1, 2, 3\}$ (no omitted reference within the loser group) and are identified jointly with the standalone Loser indicator α Loser_{*i,t*} in the regression, which absorbs the within-firm time-varying component of D1 membership. Each β_m^L therefore measures the additional loser-specific PreTOM effect in month *m* above the corresponding winner Month_m baseline. All specifications include firm and date fixed effects; standard errors are double-clustered by firm and day. Panel B is value-weighted by lagged market capitalization. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.13 52-Week-High Sort: Additional Evidence

This section reports supporting figures for the 52-week-high evidence discussed in Section 4.2.4 of the main paper. Figure IA.4 compares the share of non-paying firms across deciles under the two sorts. Figure IA.5 reports cumulative \$1 growth on long-short portfolios sorted by the two characteristics over 1980–2025.

Non-payer share by decile: 12-2 momentum vs 52-week high (1980-2025)

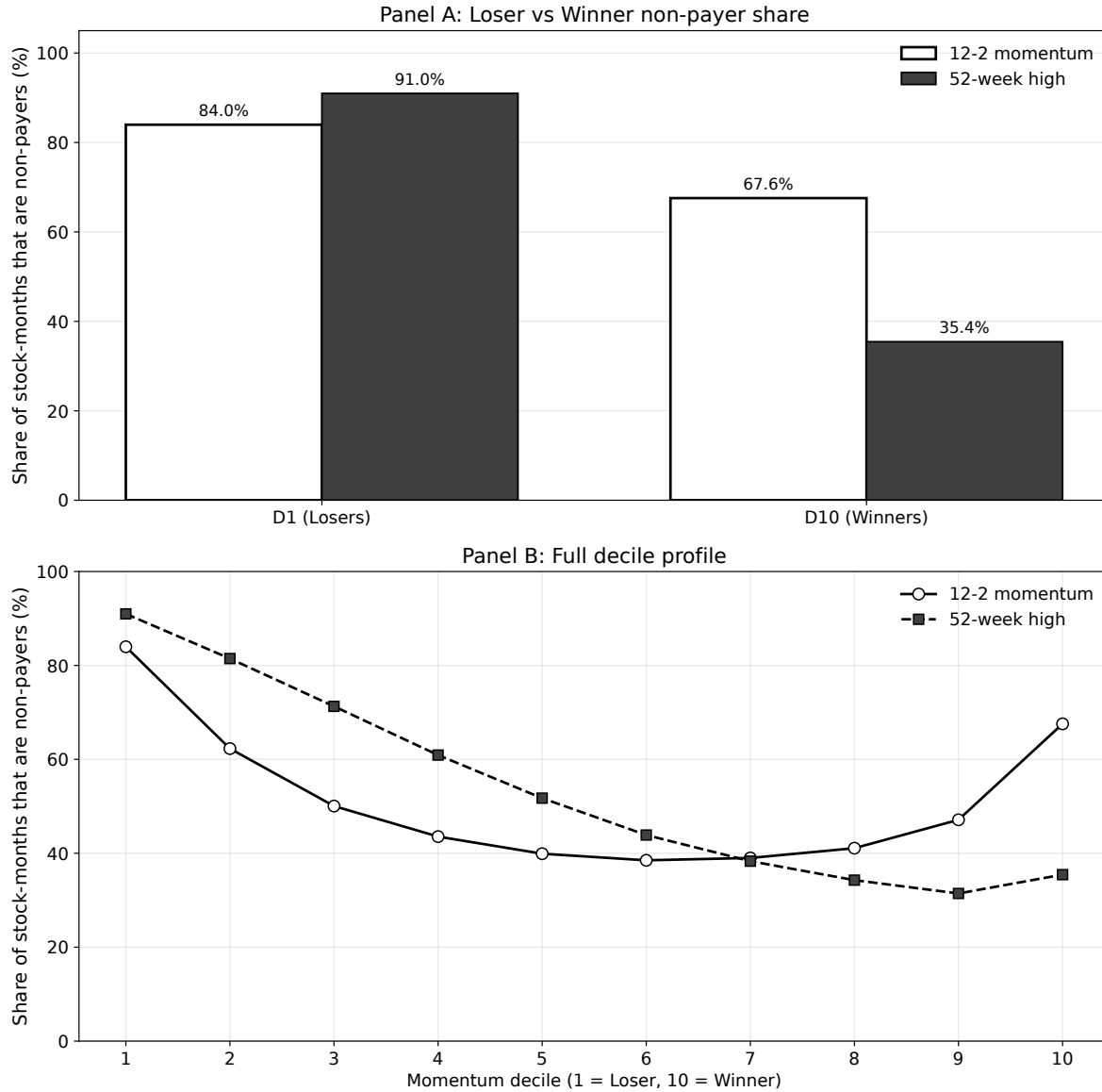


Figure IA.4: Share of non-paying stock-months by momentum decile under two sorts. Panel A: D1 (Losers) and D10 (Winners) shown as grouped bars, with 12-2 momentum (white bars with black border) and 52-week-high distance (dark gray bars). Panel B: full decile profile, 12-2 momentum (solid line with hollow markers) and 52-week-high distance (dashed line with filled markers). Both sorts are formed at the end of each month from all NYSE, AMEX, and NASDAQ common stocks in CRSP; assignments are held constant within the calendar month. Dividend-payer status is taken from CRSP distributions. Sample: 1980–2025.

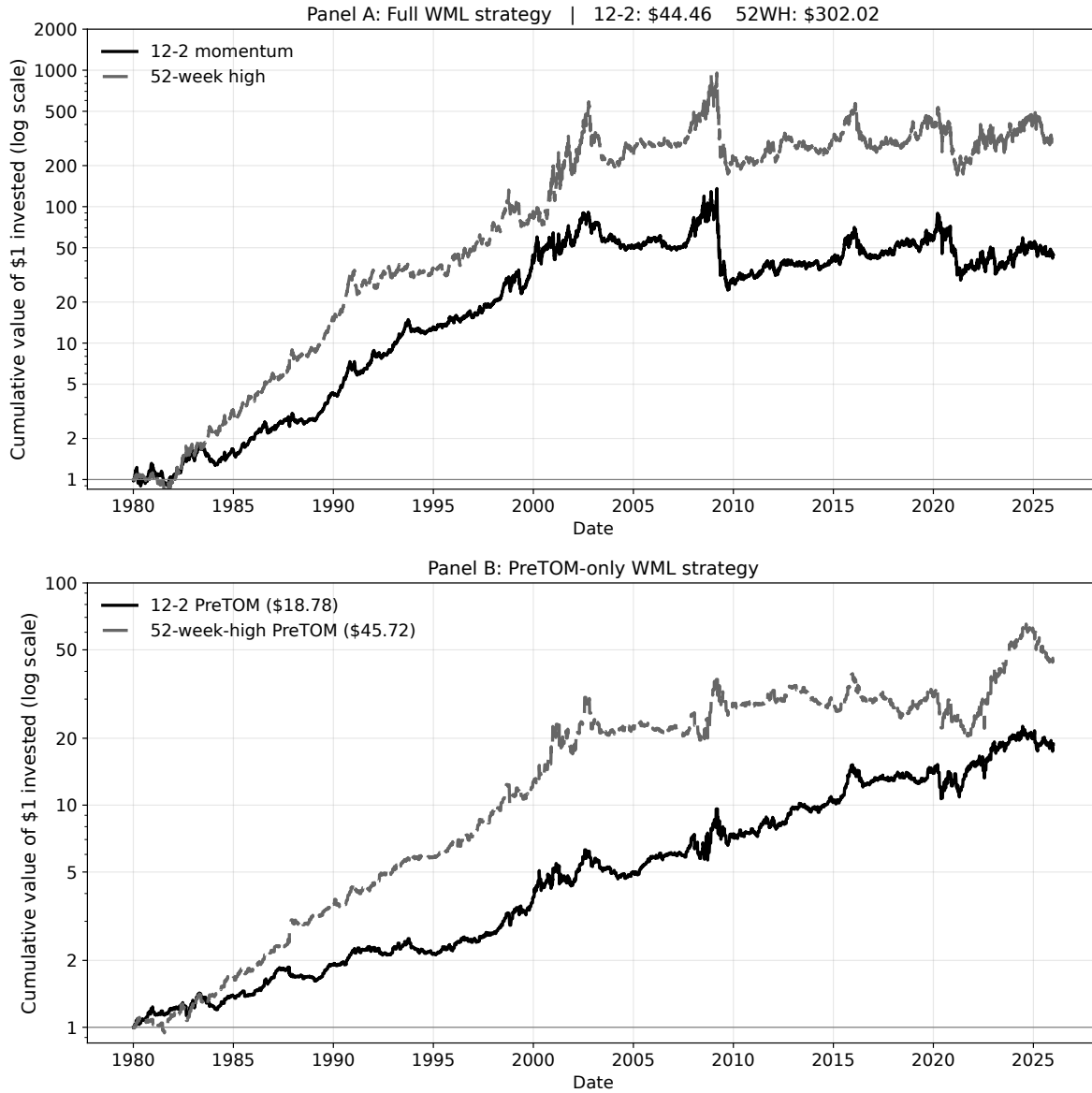


Figure IA.5: Cumulative \$1 invested in a long-short (winner – loser) portfolio over 1980–2025 under two sorts: standard 12-2 momentum (solid lines) and 52-week-high distance (dashed lines). Panel A: Full WML strategy, compounding daily WML returns every trading day. Panel B: window-conditional strategies, with PreTOM-only (in black) compounding only on days $[T-9, T-4]$ and Rest-of-month (in gray) compounding only on days outside that window; the strategy holds cash on the complementary days. Both panels are log-scaled on the y -axis with a common range. Value-weighted, fixed monthly decile assignments. End-of-sample wealth labels are shown in each legend entry.

IA.14 Liquidity Shock Panel Regression: Full Specification

This section reports the full coefficient set for the fund-month panel regression of equation (9) summarized in Table 8, Panel C of the main paper. Bold rows highlight the outflow main effect (β_1), the decile-specific outflow interactions ($\beta_{3,d}$), and the implied total outflow response by decile group.

Table IA.12: Liquidity Shock Panel Regression: Full Specification

<i>Pooled fund-month regression of equation (9); reference group = middle-decile funds (D2–D9).</i>		
Variable	Coefficient	t-stat
$D_{D1,j,y} (\beta_{1,D1})$	–1.324***	(–9.39)
$D_{D10,j,y} (\beta_{1,D10})$	+1.288***	(+9.14)
Outflow _m (β_0 , middle-decile response)	–0.339***	(–3.30)
$D_{D1,j,y} \times \text{Outflow}_m (\beta_{2,D1})$	–0.291	(–1.03)
$D_{D10,j,y} \times \text{Outflow}_m (\beta_{2,D10})$	+0.293	(+1.10)
MKT _m (γ_1)	+0.901***	(+83.75)
$D_{D1,j,y} \times \text{MKT}_m (\delta_{1,D1})$	+0.129***	(+3.36)
$D_{D10,j,y} \times \text{MKT}_m (\delta_{1,D10})$	–0.107***	(–2.92)
SMB _m (γ_2)	+0.124***	(+8.62)
$D_{D1,j,y} \times \text{SMB}_m (\delta_{2,D1})$	+0.018	(+0.28)
$D_{D10,j,y} \times \text{SMB}_m (\delta_{2,D10})$	+0.234**	(+2.55)
HML _m (γ_3)	+0.078***	(+6.00)
$D_{D1,j,y} \times \text{HML}_m (\delta_{3,D1})$	+0.013	(+0.19)
$D_{D10,j,y} \times \text{HML}_m (\delta_{3,D10})$	–0.148***	(–2.61)
<i>Implied total response to 1 pp of aggregate outflow:</i>		
Middle deciles (D2–D9): β_0	–0.339***	(–3.30)
D1 (past-losers): $\beta_0 + \beta_{2,D1}$	–0.630**	(–2.11)
D10 (past-winners): $\beta_0 + \beta_{2,D10}$	–0.046	(–0.19)
Within $R^2 = 0.695$		Obs. = 2,091,952

Notes. Estimates of equation (9), the single pooled fund-month panel regression with fund fixed effects and standard errors double-clustered by fund and month. Bold rows highlight the outflow main effect (β_0), the decile-specific outflow interactions ($\beta_{2,d}$), and the implied total outflow response by decile group, which are the coefficients summarized in Table 8, Panel B. Reference group is middle-decile funds (D2–D9). Sample: CRSP Mutual Fund Database, actively managed U.S. equity diversified funds (objective code ED, AUM $\geq$ \$10M, excluding index funds and ETFs), 1998–2025. Funds are sorted annually into deciles on prior 12-month return. The Outflow variable is coded as the magnitude of outflows (positive in outflow months). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.15 Beta Controls

This section reports the beta-controlled regressions referenced in the main text. The concern is that the Loser $\times$ PreTOM effect could be a mechanical transmission of the aggregate market’s own PreTOM underperformance through higher loser betas, rather than targeted institutional selling of losers. We use 60-month rolling CAPM betas from the Jensen et al. (2023) global factor data, demeaned within date (value-weighted for VW, equal-weighted for EW and TAQ). The full specification adds β , PreTOM $\times\beta$, Loser $\times\beta$, and Loser $\times$ PreTOM $\times\beta$ to the baseline. The triple interaction is the coefficient that would capture beta-amplification of loser selling if the channel were mechanical.

Tables IA.13, IA.14, and IA.15 show the results for value-weighted returns, equal-weighted returns, and TAQ net selling pressure respectively. Column 1 is the baseline; column 2 adds the PreTOM $\times\beta$ interaction in isolation to show its marginal contribution; column 3 is the saturated specification with the standalone β and all lower-order interactions included. Across all three, the headline Loser $\times$ PreTOM coefficient is essentially unchanged from the baseline and the triple interaction is statistically indistinguishable from zero. The PreTOM $\times\beta$ coefficient is negative and significant in returns but insignificant in TAQ selling pressure, consistent with a price-level channel rather than a flow-level channel: high-beta stocks mechanically track the market’s PreTOM drop in prices, but are not sold harder on net during PreTOM. The window effect is a flow effect on losers that lives on top of, not through, the market’s own calendar structure.

Table IA.13: Adding Stock-Level Beta as a Control to Baseline VW Panel Regression (Table 2)

	(1) Baseline	(2) + PreTOM $\times$ Beta	(3) Full
Loser	2.609** (2.04)	2.915** (2.21)	2.691** (2.33)
Loser $\times$ PreTOM	-7.151*** (-3.08)	-6.771*** (-3.08)	-7.215*** (-3.60)
Beta (demeaned)			0.935 (0.95)
PreTOM $\times$ Beta		-2.785* (-1.68)	-3.758** (-2.05)
Loser $\times$ Beta			0.499 (0.37)
Loser $\times$ PreTOM $\times$ Beta			2.648 (1.19)
Observations	53,324,145	43,368,519	43,368,519
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable matches Table 2: the daily stock return in excess of the value-weighted CRSP market return ($\text{ExRet}_{i,t} = r_{i,t} - r_t^m$), in basis points. All specifications are value-weighted (lagged market-cap), with stock and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by stock and date. Beta is the 60-month rolling CAPM beta from Jensen et al. (2023), demeaned within each date using value weights. Sample: CRSP common stocks, 1980–2025; columns 2–3 restrict to the subsample with non-missing beta. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.14: Adding Stock-Level Beta as a Control to Baseline EW Panel Regression (Table 2)

	(1) Baseline	(2) + PreTOM $\times$ Beta	(3) Full
Loser	8.022*** (11.19)	8.332*** (11.62)	8.520*** (12.97)
Loser $\times$ PreTOM	-2.550* (-1.85)	-2.566** (-1.97)	-2.472** (-2.00)
Beta (demeaned)			1.507*** (2.75)
PreTOM $\times$ Beta		-3.259*** (-3.27)	-3.997*** (-3.39)
Loser $\times$ Beta			-1.133* (-1.65)
Loser $\times$ PreTOM $\times$ Beta			0.364 (0.29)
Observations	53,324,145	43,368,519	43,368,519
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable matches Table 2: the daily stock return in excess of the value-weighted CRSP market return ($\text{ExRet}_{i,t} = r_{i,t} - r_t^m$), in basis points. All specifications are equal-weighted, with stock and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by stock and date. Beta is the 60-month rolling CAPM beta from Jensen et al. (2023), demeaned within each date. Sample: CRSP common stocks, 1980–2025; columns 2–3 restrict to the subsample with non-missing beta. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.15: Beta as Control in TAQ Selling Pressure (Table 5)

	(1) Baseline	(2) + PreTOM $\times$ Beta	(3) Full
Loser	0.0122*** (20.33)	0.0126*** (21.00)	0.0121*** (17.29)
Loser $\times$ PreTOM	0.0020** (2.50)	0.0011 (1.38)	0.0018** (2.25)
Beta (demeaned)			0.0047*** (7.83)
PreTOM $\times$ Beta		0.0018*** (4.50)	-0.0007 (-1.75)
Loser $\times$ Beta			0.0005 (0.83)
Loser $\times$ PreTOM $\times$ Beta			0.0003 (0.43)
Observations	17,861,276	15,972,786	15,972,786
Within R^2	0.0002	0.0002	0.0002

Notes. Dependent variable is daily net selling pressure (sell-initiated minus buy-initiated share of volume), TAQ data 2003–2022, approximately 17.9 million stock-day observations. All specifications include stock and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by stock and date. Beta is the 60-month rolling CAPM beta from Jensen et al. (2023), demeaned within each date. Columns 2–3 restrict to the subsample with non-missing beta. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.16 T+3 to T+2 Settlement Transition

An earlier settlement reform, the September 5, 2017 transition from T+3 to T+2 equity settlement, provides a useful calibration. The dash-for-cash mechanism predicts a one-day shift in the loser-selling boundary at this reform, from $T-5$ (the T+3-era boundary) to $T-4$ (the T+2-era boundary).

Table IA.16 reports the deadline-day DiD using the full T+3 regime as the pre-period (June 1995 – August 2017) and the T+2 era as the post-period (October 2017 – April 2024, excluding both September 2017, the regime-boundary month, and the T+1 era from May 2024 onward). The $T-4$ vs. $T-5$ DiD is -13.08 basis points ($F = 0.27$, $p = 0.60$): the point estimate is in the predicted direction (the boundary moves toward $T-4$) but is not statistically distinguishable from zero.

Table IA.16: T+3→T+2 Settlement Boundary-Day Difference-in-Differences
Panel A: 2×2 Cell Means (loser market-adjusted return, bps/day)

	Pre-T+2 (Jun 1995–Aug 2017)	Post-T+2 (Oct 2017–Apr 2024)	Diff
$T-5$	−6.99 (n = 267)	−12.08 (n = 79)	−5.09
$T-4$	−7.76 (n = 267)	−25.93 (n = 79)	−18.17
Diff ($T-4$ minus $T-5$)	−0.77	−13.85	−13.08

Panel B: DiD Regression with Early-Month Control

	Coefficient	Robust SE
θ_{T-4} : $\mathbf{1}_{[T-4]} \times \text{Post } T+2$	−20.99	(21.73)
θ_{T-5} : $\mathbf{1}_{[T-5]} \times \text{Post } T+2$	−7.92	(21.34)
$\theta_{T-4} - \theta_{T-5}$ (boundary shift)	−13.08	(25.23)
$F = 0.27, p = 0.604$		

Notes: Boundary-day difference-in-differences test of the T+3 to T+2 settlement transition (September 5, 2017). Dependent variable: daily market-adjusted VW loser-portfolio return (basis points). Treated days are $T-5$ and $T-4$ (the pre- and post-reform boundary days under the linearity prediction); control days are early-month $[T+5, T+8]$. Pre-period: June 1995 (start of the T+3 regulatory regime) to August 2017; post-period: October 2017 to April 2024 (excluding both September 2017, the regime-boundary month, and the T+1 era from May 2024 onward). The headline test $\theta_{T-4} - \theta_{T-5} < 0$ is in the predicted direction (the boundary moves from $T-5$ toward $T-4$) but not statistically distinguishable from zero. Robust standard errors in parentheses. By construction the Panel A $T-4$ minus $T-5$ diff equals the Panel B $\theta_{T-4} - \theta_{T-5}$; the early-month control absorbs the common post-period drift but cancels out of the boundary contrast.

IA.17 Winner Window Effect

The asymmetry of the PreTOM effect, mostly loser-driven, is confirmed by running the baseline regression on winners instead of losers. Table IA.17 reports the winner analogue of the baseline Loser $\times$ PreTOM regression. Neither the Winner main effect nor the Winner $\times$ PreTOM interaction is statistically distinguishable from zero in the value-weighted specification, confirming that winners show no differential PreTOM concentration.

Table IA.17: Winner Window Effect on Daily Returns

	EW	VW
Winner	−0.23*** (−3.29)	0.04 (0.40)
Winner $\times$ PreTOM	−1.27 (−0.82)	1.55 (0.76)
Fixed effects	Firm, Date	Firm, Date
Observations	53,324,149	53,324,149
Within R^2	0.0000	0.0000

Notes. Dependent variable is the daily stock return in excess of the value-weighted CRSP market ($r_{i,t} - r_t^m$), in basis points. Value-weighted (VW) regressions use lagged market-capitalization weights. t -statistics (in parentheses) are computed from standard errors double-clustered by firm and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: CRSP common stocks, 1980–2025.

IA.18 S&P 500 Membership Robustness

A natural concern is that the PreTOM effect is concentrated among small-cap or thinly traded stocks. The effect is also present among S&P 500 constituents: the Loser $\times$ PreTOM coefficient is -8.77 ($t = -2.71$) among S&P 500 stocks and -4.62 ($t = -2.53$) among non-constituents, with no statistically significant difference between the two groups (triple interaction $t = -0.82$). The effect is not confined to small or illiquid stocks.

Table IA.18: Window Effect and S&P 500 Membership

	Non-S&P 500 (1)	S&P 500 (2)	Full Sample (3)
Loser	1.806* (1.84)	4.018** (2.23)	2.429** (2.20)
Loser $\times$ PreTOM	-4.622^{**} (-2.53)	-8.769^{***} (-2.71)	-5.916^{**} (-2.56)
Loser $\times$ S&P 500			0.408 (0.25)
Loser $\times$ PreTOM $\times$ S&P 500			-2.520 (-0.82)
Observations	47,863,849	5,460,280	53,324,145
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable is the daily stock return in excess of the value-weighted CRSP market ($r_{i,t} - r_t^m$), in basis points. Value-weighted using lagged market-capitalization weights. Columns (1) and (2) restrict the sample to non-S&P 500 and S&P 500 constituent stocks, respectively; columns (1)+(2) do not exactly sum to column (3) because the S&P 500 membership flag is missing on a small number of stock-days (16 obs), which are dropped from the split samples but retained in the full-sample interaction specification. Column (3) estimates the full sample with an S&P 500 interaction. All specifications include firm and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by firm and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: CRSP common stocks, 1980–2025.

IA.19 Long-Horizon Reversion: Identification of the PreTOM-Specific Component

The short-horizon Post-window narrowing documented in the main paper shows that the loser-vs-non-loser gap that opens during PreTOM closes within the surrounding seven days. The same mechanism implies a longer-horizon prediction: cumulative pressure from past PreTOM windows should mean-revert at horizons of several months to two years, while cumulative returns over non-PreTOM days, which reflect information and fundamentals rather than transitory flow pressure, should not. The asymmetric prediction tightens the test: identification comes from the contrast between the two predictors rather than the absolute level of either.

We focus on the D1 loser decile. Panel A of Table IA.19 reports daily time-series regressions on the value-weighted D1 portfolio:

$$r_t - r_t^m = \alpha + \beta_1 \text{CumPT}_t + \beta_2 \text{CumNPT}_t + \beta_3 \text{CumRec}_t + \beta_4 \text{PreTOM}_t + \beta_5 (r_{t-1} - r_{t-1}^m) + \varepsilon_t, \quad (\text{IA.1})$$

where $\text{CumPT}_t = \sum_{s=t-480}^{t-61} (r_s - r_s^m) \cdot \mathbf{1}\{s \in \text{PreTOM}\}$ sums distant past market-adjusted returns on PreTOM days only; $\text{CumNPT}_t = \sum_{s=t-480}^{t-61} (r_s - r_s^m) \cdot \mathbf{1}\{s \notin \text{PreTOM}\}$ sums distant past market-adjusted returns on non-PreTOM days; and $\text{CumRec}_t = \sum_{s=t-60}^{t-1} (r_s - r_s^m)$ is the recent 60-day cumulative market-adjusted return. The distant window runs from 60 to 480 trading days prior (approximately three to twenty-four months). The regression is estimated with Newey–West standard errors at 21 lags. The prediction is $\beta_1 < 0$ and $\beta_2 \approx 0$, with the Wald test of $\beta_1 = \beta_2$ rejecting symmetric reversion.

The pattern holds in the standard D1 portfolio ($\beta_1 = -0.0033$, $t = -2.32$; Wald $t = -2.41$) and strengthens when the portfolio is restricted to stocks with persistent past D1-PreTOM exposure. Conditioning today’s portfolio on stocks that were in D1 on at least three past PreTOM days (out of approximately 120 in the distant window) drives β_1 to -0.0043 ($t = -3.06$) and the Wald statistic to $t = -3.04$, with effectively no loss in portfolio size (575 vs. ~ 500 stocks/day; only one percent of D1 market capitalization is removed). Results are stable across thresholds from ≥ 1 through ≥ 30 ; we report ≥ 3 as the primary specification because it removes stocks with essentially zero past D1-PreTOM exposure while preserving 99 percent of D1 market capitalization. Within the ≥ 3 universe, CumPT_t has a time-series

standard deviation of 1,779 basis points; the implied cross-sectional spread between $\pm 1\sigma$ tails is 15.4 basis points per day, roughly 39 percentage points annualized. Conditioning on past D1 status also drives β_2 to zero (from +0.0002 to -0.0001), removing the trace of conventional medium-horizon momentum that contaminated the unfiltered specification (Jegadeesh and Titman, 1993).

Panel B identifies the same reversion within-firm, with firm and day fixed effects absorbing stock-specific means and aggregate day-level shocks. The key step is to decompose each stock's past PreTOM returns by its own past D1 status: $\text{CumPT}_{i,t}^{\text{PL}}$ sums past PreTOM returns on days when stock i was itself in D1, and $\text{CumPT}_{i,t}^{\text{PNL}}$ sums past PreTOM returns from periods when the stock was not. Analogous quantities $\text{CumNPT}_{i,t}^{\text{PL}}$ and $\text{CumNPT}_{i,t}^{\text{PNL}}$ decompose past non-PreTOM returns. The panel regression is

$$\begin{aligned} r_{i,t} - r_t^m = & \alpha_i + \delta_t + b_1 \text{CumPT}_{i,t}^{\text{PL}} + b_2 \text{CumPT}_{i,t}^{\text{PNL}} \\ & + b_3 \text{CumNPT}_{i,t}^{\text{PL}} + b_4 \text{CumNPT}_{i,t}^{\text{PNL}} + b_5 \text{CumRec}_{i,t} + b_6 \text{Loser}_{i,t} + \varepsilon_{i,t}, \end{aligned} \quad (\text{IA.2})$$

estimated on all momentum deciles (43.5 million stock-days), value-weighted by lagged market capitalization, with standard errors two-way clustered by firm and day.

Past PreTOM returns accumulated while the stock was in D1 ($b_1 = -0.00111$, $t = -4.99$) revert about three times more strongly than past PreTOM returns accumulated while the stock was outside D1 ($b_2 = -0.00038$, $t = -2.65$), with the Wald test of equality rejecting at $p = 0.0006$. Past non-PreTOM returns show no equivalent asymmetry: b_3 is indistinguishable from zero in D1 and b_4 is marginal outside D1. The marginal b_4 is consistent with a small generic long-horizon reversal in non-loser stocks that firm fixed effects do not fully absorb; its magnitude is about 60 percent of b_2 .

If the negative coefficient on CumPT^{PL} reflected generic mean-reversion of persistent underperformers, the analogous $\text{CumNPT}^{\text{PL}}$ should also be negative and large; it is not. The reversion loads specifically on the component of past returns that accumulated during PreTOM windows, when flow pressure was active. Standard persistent-regressor concerns (Stambaugh, 1999; Ferson et al., 2003) affect both predictors similarly given their parallel construction and difference out in the Wald test.

Panel B identifies whether each predictor matters; it does not say where the reversion shows up. Since the mechanism in the main paper concentrates on stocks currently classified

as losers, we ask whether reversion of past D1-PreTOM accumulation realizes when stock i remains in D1 today, when it has exited, or both. We augment equation (IA.2) by interacting all four cumulative-return predictors with $\text{Loser}_{i,t}$:

$$\begin{aligned} r_{i,t} - r_t^m = & \alpha_i + \delta_t + \sum_{k=1}^4 b_k X_{i,t}^{(k)} + \sum_{k=1}^4 b'_k X_{i,t}^{(k)} \times \text{Loser}_{i,t} \\ & + b_5 \text{CumRec}_{i,t} + b_6 \text{Loser}_{i,t} + b_7 \text{PreTOM}_t + \varepsilon_{i,t}, \end{aligned} \quad (\text{IA.3})$$

where $X_{i,t}^{(1)} = \text{CumPT}_{i,t}^{\text{PL}}$, $X_{i,t}^{(2)} = \text{CumPT}_{i,t}^{\text{PNL}}$, $X_{i,t}^{(3)} = \text{CumNPT}_{i,t}^{\text{PL}}$, and $X_{i,t}^{(4)} = \text{CumNPT}_{i,t}^{\text{PNL}}$. The main coefficients b_k now capture reversion on stock-days where i is currently outside D1; the linear combinations $b_k + b'_k$ capture reversion on stock-days where i remains in D1. Panel C of Table IA.19 reports the results.

Among stocks currently outside D1, only past D1-PreTOM accumulation predicts future returns ($b_1 = -0.000713$, $t = -3.61$); the other three components are indistinguishable from zero. This is the post-exit prediction the main paper's narrative requires: once a stock leaves D1, only its history of having been pressed during PreTOM, and only on PreTOM days while it was in D1, predicts its subsequent path. In-D1 stock-days show stronger reversion across all four components ($b_k + b'_k$ negative and statistically significant for $k = 1, \dots, 4$), but the all-negative pattern combines reversion with the mechanical contrarian property of the loser portfolio: stocks classified as losers today are stocks that have just fallen, and the sample is dense in further short-horizon mean-reversion of recent losses (Lehmann, 1990; Jegadeesh and Titman, 1995). The diagnostic test is therefore the out-of-D1 column, where this density is absent. There, CumPT^{PL} stands alone.

Table IA.19: Long-Horizon Reversion of PreTOM Loser Pressure

<i>Panel A: Portfolio Time Series (D1, value-weighted)</i>				
Portfolio	Avg stocks/day	β_1 : CumPT	β_2 : CumNPT	Wald: $\beta_1=\beta_2$
Standard D1 (no filter)	~ 500	-0.0033^{**} (-2.32)	$+0.0002$ ($+0.30$)	$t = -2.41^{**}$
Persistent ≥ 3 (primary)	575	-0.0043^{***} (-3.06)	-0.0001 (-0.06)	$t = -3.04^{***}$
Persistent ≥ 30	400	-0.0036^{***} (-2.63)	-0.0000 (-0.03)	$t = -2.43^{**}$
Persistent ≥ 110	35	-0.0019 (-1.44)	-0.0006 (-0.96)	$t = -0.90$

Notes (Panel A). Daily time-series regressions of value-weighted D1 portfolio returns on rolling cumulative past returns over the distant window $[t-480, t-60]$. CumPT_{*t*} sums returns on past PreTOM days ($T-9$ through $T-4$); CumNPT_{*t*} sums returns on past non-PreTOM days. Specifications include a PreTOM_{*t*} dummy and the portfolio's own one-day lagged return as additional controls. Newey-West standard errors, 21 lags. The persistent- $\geq k$ portfolios augment today's D1 universe with stocks that were in D1 on at least k PreTOM days within the distant window $[t-480, t-60]$ (out of approximately 120 such days), even if they have since exited D1. That is why the persistent samples can have a larger average stocks/day than the unfiltered "Standard D1" panel: they include past loser cohorts in addition to today's D1. Panels B and C continue overleaf. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.19: Long-Horizon Reversion of PreTOM Loser Pressure (continued)

<i>Panel B: Stock-Level Panel (all deciles, firm + date FE, VW)</i>			
Variable	Estimate (<i>t</i> -stat)		
b_1 : CumPT $_{i,t}^{\text{PL}}$ (past PreTOM while in D1)	−0.00111***	(−4.99)	
b_2 : CumPT $_{i,t}^{\text{PNL}}$ (past PreTOM while not in D1)	−0.00038***	(−2.65)	
b_3 : CumNPT $_{i,t}^{\text{PL}}$ (past non-PreTOM while in D1)	−0.00013	(−0.79)	
b_4 : CumNPT $_{i,t}^{\text{PNL}}$ (past non-PreTOM while not in D1)	−0.00023*	(−1.88)	
b_5 : CumRec $_{i,t}$	−0.00253***	(−7.14)	
b_6 : Loser $_{i,t}$	−2.16**	(−2.24)	
Wald test: $b_1 = b_2$	$F = 11.85, p = 0.0006$		
Difference ($b_1 - b_2$)	−0.00073***	(−3.44)	
Observations (stock-days)	43,463,404		
<i>Panel C: Reversion by Current D1 Status (all four predictors interacted with Loser$_{i,t}$)</i>			
Past-return component	Out of D1 (main: b_k)	In D1 (main + int.: $b_k + b'_k$)	× Loser (b'_k)
CumPT $_{i,t}^{\text{PL}}$	−0.000713*** (−3.61)	−0.001569*** (−4.33)	−0.000857** (−2.38)
CumPT $_{i,t}^{\text{PNL}}$	−0.000211 (−1.50)	−0.001484*** (−3.85)	−0.001273*** (−3.46)
CumNPT $_{i,t}^{\text{PL}}$	+0.000116 (+0.69)	−0.000574** (−2.13)	−0.000690*** (−2.59)
CumNPT $_{i,t}^{\text{PNL}}$	−0.000161 (−1.36)	−0.000720*** (−2.67)	−0.000559** (−2.27)
Wald test: $b'_1 = 0$ (interaction on CumPT $^{\text{PL}}$)	$F = 5.67, p = 0.017$		
Joint test: $b'_1 = b'_2 = b'_3 = b'_4 = 0$	$F = 4.92, p = 0.0006$		
Test $b_1 = 0$ (out-of-D1 reversion of CumPT $^{\text{PL}}$)	$F = 13.02, p = 0.0003$		
Test $b_1 + b'_1 = 0$ (in-D1 reversion of CumPT $^{\text{PL}}$)	$F = 18.79, p < 0.001$		
Observations (stock-days)	43,463,404		
Within R^2	0.2304		

Notes (Panels B and C). Predictors are rolling cumulative returns over the distant window $[t-480, t-60]$, decomposed by whether stock i was itself in D1 on each past day: CumPT $^{\text{PL}}$ (CumPT $^{\text{PNL}}$) sums past PreTOM returns when stock i was (was not) in D1 on day s , and analogously for non-PreTOM days. CumRec $_t$ sums returns over the recent $[t-60, t-1]$ window. Panel B reports the panel regression of equation (IA.2); all deciles, firm and date FE, VW by lagged mcap, errors clustered by firm and date. Panel C augments by interacting all four cumulative-return predictors with Loser $_{i,t}$; “Out of D1” columns report the main coefficient b_k , “In D1” columns report $b_k + b'_k$, rightmost column reports the interaction b'_k . Additionally controls for CumRec $_{i,t}$, Loser $_{i,t}$, and PreTOM $_t$. NYSE breakpoints (CRSP common stocks); deciles fixed within each calendar month. Sample 1980–2025. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.20 Additional Figures

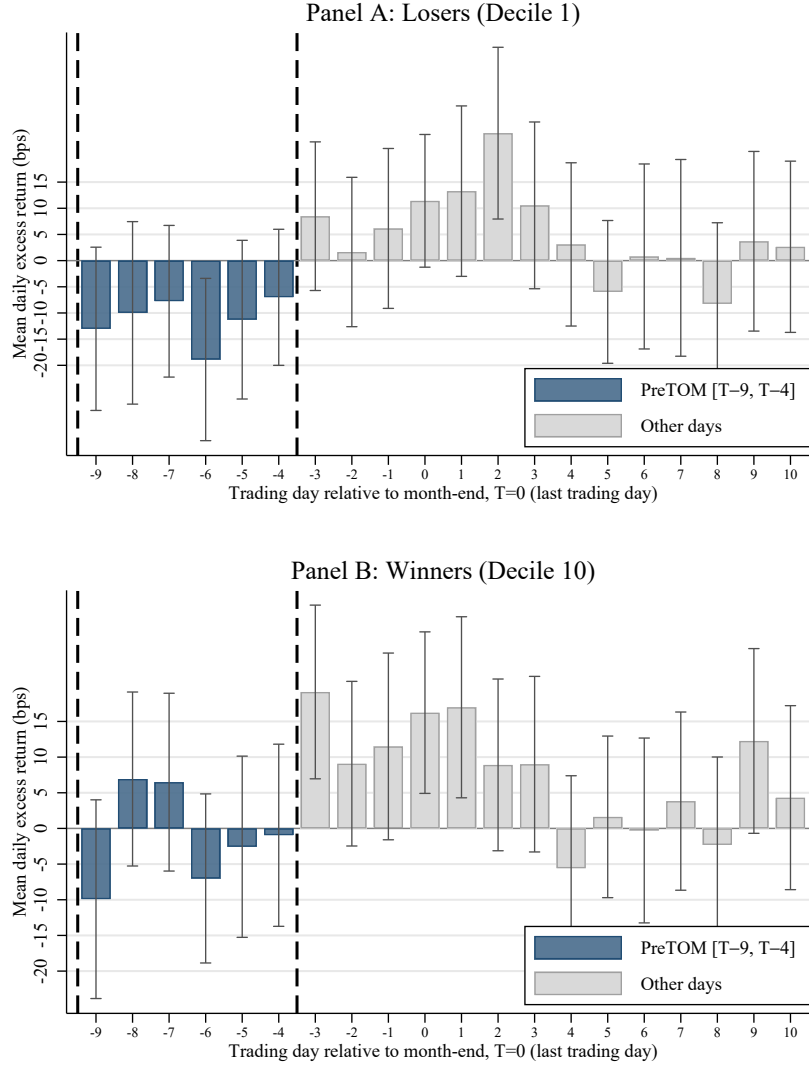


Figure IA.6: Mean daily excess returns ($r_{i,t} - r_t^f$, bps) by trading day T ($T=0$ is month-end). Panel A: D1 losers. Panel B: D10 winners. PreTOM bars ($[T-9, T-4]$) dark blue, other days light gray. For each trading day T , daily returns are compounded to monthly returns within T , and the 95% confidence interval is the cross-monthly mean $\pm 1.96 \times$ the cross-monthly standard error. VW, fixed monthly deciles. 1980–2025.

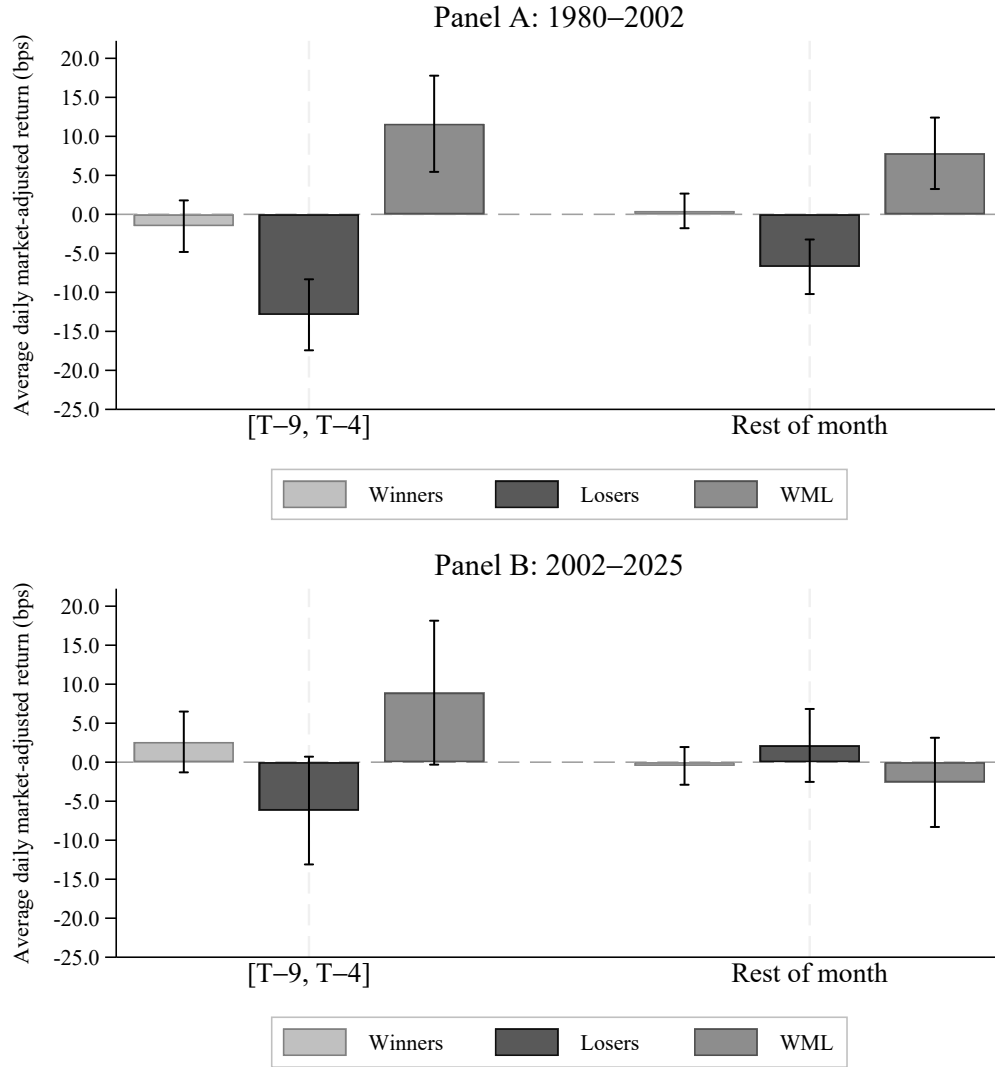


Figure IA.7: Average daily market-adjusted returns (basis points) of momentum winners, losers, and WML during PreTOM ($T-9$ to $T-4$) versus Rest, by subperiod. Market-adjusted returns are $r_i - r_{mkt}$ for winners and losers; WML is unchanged. Bars show time-series means; whiskers denote 95% confidence intervals. Data are daily value-weighted returns using fixed monthly decile assignments from CRSP.

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